



Learning-based model augmentation with LFRs[☆]

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ABSTRACT

Nonlinear system identification (NL-SI) has proven to be effective in obtaining accurate models for highly complex systems. In particular, recent encoder-based methods for *artificial neural network state-space* (ANN-SS) models have achieved state-of-the-art performance on various benchmarks, while offering consistency and computational efficiency. Inclusion of prior knowledge of the system can be exploited to increase (i) estimation speed, (ii) accuracy, and (iii) interpretability of the resulting models. This paper proposes an encoder-based model augmentation method that incorporates prior knowledge from *first-principles* (FP) models. We introduce a novel *linear-fractional-representation* (LFR) model structure that allows for the unified representation of various augmentation structures including the ones that are commonly used in the literature, and an identification algorithm for estimating the proposed structure together with appropriate initialization methods. The performance and generalization capabilities of the proposed method are demonstrated in the identification of a hardening mass-spring-damper system simulation case study and the Bouc-Wen nonlinear system identification benchmark.

1. Introduction

As systems used in control become increasingly complex and performance requirements increase, the demand for accurate nonlinear models that capture complicated behaviors is rapidly growing. Obtaining these models through *first-principles* (FP) based methods is becoming too costly and often infeasible due to the surge of system complexity and difficult to model phenomena for which no reliable FP descriptions exist. To overcome these issues, *nonlinear system identification* (NL-SI) methods can be used to estimate models based on measured data from the system (Schoukens & Ljung, 2019). In particular, the estimation of black-box models combined with *artificial neural networks* (ANNs) has been able to achieve unprecedented accuracy. In control, ANN *state-space* (SS) models have been used to cope with high-order systems and capture complex nonlinear dynamic effects (Schoukens, 2021; Suykens, De Moor, & Vandewalle, 1995). Recent encoder-based methods for ANN-SS models (Beintema, Schoukens & Tóth, 2023; Beintema et al., 2023; Forgione & Piga, 2021) have shown state-of-the-art performance on a wide range of benchmark problems, while providing consistency guarantees and computational efficiency.

Although the identified models obtained with black-box methods may be accurate, they are not based on physical laws and are difficult

to interpret. This forms a substantial roadblock to the use of these models in industry, where interpretable models are preferred for the design process (Chioua (moderator), 2021; Ljung, 2010). Furthermore, black-box methods do not consider the prior knowledge available for the system, such as (approximate) FP models that have been created in the past. As a result, black-box methods will discover already known dynamics, which is an apparent waste of prior knowledge and modeling resources and often results in excessive amount of estimation time.

Physics-based learning methods can include prior information of the system in the NL-SI algorithm in a variety of ways (Daw et al., 2022; Pathak et al., 2018; Psychogios & Ungar, 1992; Willard et al., 2022), resulting in (i) faster estimation and (ii) better accuracy on unseen data than black-box methods. Model augmentation is a promising approach in which prior information is included by combining an FP-based model with flexible function approximators in a variety of different model augmentation structures, such as parallel (Lai, Mylonas, Nagarajaiah, & Chatzi, 2021; Retzler et al., 2024; Sun et al., 2020) and series (Bohlin, 2006; Götte & Timmermann, 2022; Groote et al., 2022; Retzler et al., 2024; Shah et al., 2022) interconnections. Such augmentation structures can result in (iii) more interpretable models.

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The model augmentation methods mentioned above produce models with benefits (i)–(iii) compared to ANN-SS identification methods; however, they often require full-state measurements (Götte & Timmermann, 2022; Groote et al., 2022; Shah et al., 2022; Sun et al., 2020) or an estimation of the states (Bohlin, 2006). In practice, often only input–output data is available. Secondly, existing model augmentation structures only consider a limited number of interconnections between the learning components and the baseline model. Such interconnections might be necessary for accurate augmentation of the state transition function or adding missing dynamic (states) to the baseline model. A more flexible model augmentation structure and algorithms capable of reliable model learning with it would be required to consider such interconnections.

The encoder-based ANN-SS methods can handle input–output data efficiently and have shown state-of-the-art results in deep learning-based system identification (Beintema, Schoukens et al., 2023; Beintema et al., 2023; Forgiione & Piga, 2021). Hence, a combination of model augmentation and encoder-based methods seems promising. Despite this, the extension of encoder-based methods to model augmentation scenarios has been limited to a case study with promising results (Retzler et al., 2024).

We propose a model-augmentation-based identification method building on recent results in NL-SI in terms of the *Sub-Space Encoder Network* (SUBNET) (Beintema, Schoukens et al., 2023; Beintema, Tóth, & Schoukens, 2021; Beintema et al., 2023). Our contribution in this paper is the adaption of SUBNET to the model-augmentation identification problem. To achieve this, we propose a novel *linear-fractional-representation* (LFR) (Redheffer, 1960) based model augmentation structure that flexibly combines the baseline model with a black-box learning component. In the LFR model structure, an interconnection matrix governs the signals between the baseline model and the augmentation. By shaping this interconnection matrix, we can represent a wide range of model augmentation structures proposed in the literature (Bolderman, 2024; Daw et al., 2022; Götte & Timmermann, 2022; Schoukens & Tóth, 2020; Sohlberg et al., 2008; Thompson & Kramer, 1994) and a wide variety of structures beyond them, resulting in an unifying model structure represented in LFR form. The LFR model structure also allows for the inclusion of additional dynamical components, in terms of states of the learning component, beyond the baseline model dynamics. Furthermore, we also consider joint identification of the baseline model parameters and the learning component parameters and handle the problem of overparameterization through regularization (Bolderman, 2024). The contributions are summarized as follows:

- We propose a LFR-based model augmentation structure unifying existing model augmentation structures;
- We extend the LFR form for dynamic model augmentation to better capture missing dynamics;
- We provide a system identification algorithm capable of joint estimation of the baseline model parameters and the ANN parameterization of the learning components in the proposed LFR-based model augmentation structure.

In the remainder of the paper, first the NL-SI problem is introduced with baseline models in Section 2, followed by the introduction of the LFR-based model augmentation structure, dynamic augmentation, and system identification algorithm in Section 3. A hardening *mass-spring-damper* (MSD) system based simulation example and the Bouc-Wen hysteresis benchmark (Schoukens & Noël, 2017) are used to demonstrate the performance of the proposed identification method and compare it to black-box ANN-SS methods in Sections 4 and 5. Conclusions are given in Section 6.

2. Identification problem

We consider the data-generating system given by the *discrete-time* (DT) nonlinear representation

$$x_{k+1} = f(x_k, u_k), \quad (1a)$$

$$y_k = h(x_k) + e_k, \quad (1b)$$

where $x_k \in \mathbb{R}^{n_x}$ is the state, $u_k \in \mathbb{R}^{n_u}$ is the input, $y_k \in \mathbb{R}^{n_y}$ is the output signal of the system at time moment $k \in \mathbb{Z}$ with e_k an i.i.d., possibly colored, noise process with finite variance representing measurement noise. The state-transition function $f: \mathbb{R}^{n_x} \times \mathbb{R}^{n_u} \rightarrow \mathbb{R}^{n_x}$ and the output function $h: \mathbb{R}^{n_x} \rightarrow \mathbb{R}^{n_y}$ are such that the solutions of (1) are forward complete.

A baseline model for the system (1) can, for example, be obtained by FP methods. The baseline model is given as

$$\tilde{x}_{k+1} = f_{\text{base}}(\theta_{\text{base}}, \tilde{x}_k, u_k), \quad (2a)$$

$$\tilde{y}_k = h_{\text{base}}(\theta_{\text{base}}, \tilde{x}_k, u_k), \quad (2b)$$

where $\tilde{x}_k \in \mathbb{R}^{n_{\tilde{x}}}$ is the baseline model state, $\tilde{y}_k \in \mathbb{R}^{n_{\tilde{y}}}$ the baseline model output, and f_{base} and h_{base} are the baseline model functions for state transition and output readout, respectively, with parameters θ_{base} . These functions are jointly denoted as ϕ_{base} . These baseline models can be good approximations of the system (1). However, as motivated in Section 1, obtaining accurate baseline models is costly and often infeasible. Hence, often only approximate baseline models with, for example, incomplete dynamics and approximate parameters θ_{base} are available. We then augment these baseline models with learning components through a model augmentation method.

In model augmentation, our aim is to represent (1) by a model that combines the baseline model (2) with learning components. Using a data sequence $D_N = \{(y_k, u_k)\}_{k=1}^N$ generated by (1), the aim is to identify a DT model augmentation structure of the form

$$\hat{x}_{k+1} = f_{\text{base}} \star f_{\text{aug}}(\theta_{\text{base}}, \theta_{\text{aug}}, \hat{x}_k, u_k), \quad (3a)$$

$$\hat{y}_k = h_{\text{base}} \star h_{\text{aug}}(\theta_{\text{base}}, \theta_{\text{aug}}, \hat{x}_k, u_k), \quad (3b)$$

where $\hat{x}_k \in \mathbb{R}^{n_{\hat{x}}}$ is the model state, $\hat{y}_k \in \mathbb{R}^{n_{\hat{y}}}$ is the model output, and $\star$ is an interconnection between the baseline model and the learning components f_{aug} and h_{aug} with the parameters θ_{aug} . In the literature, a variety of special cases of this general model augmentation structure are presented, such as static parallel (Lai et al., 2021; Retzler et al., 2024; Sun et al., 2020) and static series (Götte & Timmermann, 2022; Groote et al., 2022; Retzler et al., 2024; Shah et al., 2022) structures, as shown in Table 1, which have been developed in the fields mechanistic, process industry and structural modeling. We consider these model structures as static augmentations, as they do not add additional states beyond the baseline model to capture dynamics missing from the baseline model, such as flexible modes, actuator dynamics, etc., compared to a baseline rigid-body model. Dynamic augmentation structures such as (Bohlin, 2006) have also been studied. In this case, the model state $\hat{x}$ is $[\bar{x}^T \ \bar{x}^T]^T$, where $\bar{x}$ is the additional state added for the dynamic learning component.

These model augmentation structures found in the literature are limited in the interconnections considered between the baseline model and the learning components. For example, we consider novel dynamic augmentations in which the learning components take the model state $\hat{x}$ as input, and their output affects the entire model state $\hat{x}$ as shown in Table 1. This dynamic augmentation, as well as other augmentation structures, could be represented by a more flexible model augmentation structure representation of (3).

An efficient model augmentation structure should be able to represent all the cases outlined in Table 1 and even a wide variety of model augmentation arrangements beyond them. Such a flexible model structure will also require an algorithm capable of reliable joint estimation of the baseline θ_{base} and augmentation parameters θ_{aug} . To address

Table 1
Classes of model augmentation structures.

	Static parallel	Static series
$\tilde{x}_{k+1}$	$f_{\text{base}}(\tilde{x}_k, u_k) + f_{\text{aug}}(\tilde{x}_k, u_k)$	$f_{\text{aug}}(f_{\text{base}}(\tilde{x}_k, u_k))$
$\tilde{x}_{k+1}$	—	—
	Dynamic parallel	Dynamic series
$\tilde{x}_{k+1}$	$f_{\text{base}}(\tilde{x}_k, u_k) + f_{\text{aug}}(\tilde{x}_k, \tilde{x}_k, u_k)$	$f_{\text{aug}}(f_{\text{base}}(\tilde{x}_k, u_k))$
$\tilde{x}_{k+1}$	$g_{\text{aug}}(\tilde{x}_k, \tilde{x}_k, u_k)$	$g_{\text{aug}}(\tilde{x}_k, \tilde{x}_k, u_k)$

these limitations, we propose the adaptation of SUBNET (Beintema, Schoukens et al., 2023; Beintema et al., 2021, 2023) to the model augmentation setting and a novel LFR-based model augmentation structure. We consider ANNs¹ as learning components.

3. Method

3.1. LFR-based augmentation structure

A key property of LFR representations is that the elementary operations of summation, multiplication, and inversion allow manipulations with LFRs, and an LFR of an LFR is also an LFR (Redheffer, 1960). The flexibility of this representation has made it popular in the field of robust control (Junnarkar et al., 2022; Zhou, Doyle, & Glover, 1996) and linear parameter-varying-control (Schoukens & Tóth, 2018; Tóth, 2010). Furthermore, an LFR can also include nonlinear components in the interconnections (Veenman, Scherer, & Kőroğlu, 2016), which has made LFRs useful for black-box nonlinear system representations (Schoukens & Tóth, 2020; Shakib et al., 2022) and implicit learning (El Ghaoui et al., 2021). Recent results have also shown that stability properties can be enforced on these black-box LFRs at the cost of some representation capability (Frank et al., 2022; Revay et al., 2020, 2021). In this work, we will use LFRs to include a baseline model of the system in a rather flexible arrangement w.r.t. the learning component.

We augment the baseline model (2) with learning components in terms of parameterized functions, jointly denoted by ϕ_{aug} . The proposed LFR-based augmentation structure combines ϕ_{base} and ϕ_{aug} as shown in Fig. 1 and can be written as

$$\begin{bmatrix} \hat{x}_{k+1} \\ \hat{y}_k \\ z_{1,k} \\ z_{2,k} \end{bmatrix} = \underbrace{\begin{bmatrix} S_{xx} & S_{xu} & S_{xw_1} & S_{xw_2} \\ S_{yx} & S_{yu} & S_{yw_1} & S_{yw_2} \\ S_{z_1x} & S_{z_1u} & S_{z_1w_1} & S_{z_1w_2} \\ S_{z_2x} & S_{z_2u} & S_{z_2w_1} & S_{z_2w_2} \end{bmatrix}}_S \begin{bmatrix} \hat{x}_k \\ u_k \\ w_{1,k} \\ w_{2,k} \end{bmatrix} \quad (4a)$$

$$w_{1,k} = \phi_{\text{base}}(z_{1,k}) = \begin{bmatrix} f_{\text{base}}(\theta_{\text{base}}, z_{1,k}) \\ h_{\text{base}}(\theta_{\text{base}}, z_{1,k}) \end{bmatrix} \quad (4b)$$

$$w_{2,k} = \phi_{\text{aug}}(z_{2,k}), \quad (4c)$$

where $S \in \mathbb{R}^{n \times m}$ is the interconnection matrix and S are selection matrices of appropriate size.

3.2. Conditions on the interconnection matrix

The interconnection matrix S can be fixed or parameterized. In this work, we consider fixed interconnection matrices S . By further choosing ϕ_{aug} formulations, the model augmentation structures in Table 1

¹ Consider that each hidden layer is composed of m activation functions $\rho: \mathbb{R} \rightarrow \mathbb{R}$ in the form of $z_{i,j} = \rho(\sum_{l=1}^{m-1} \theta_{w,i,j,l} z_{i-1,l} + \theta_{b,i,j})$ where $z_i = \text{col}(z_{i,1}, \dots, z_{i,m})$ is the latent variable representing the output of layer $1 \leq i \leq q$. Here, $\text{col}(\cdot)$ denotes the composition of a column vector. For a f_θ with q hidden layers and linear input and output layers, this means $f_\theta(\hat{x}_k, u_k) = \theta_{w,q+1} z_q(k) + \theta_{b,q+1}$ and $z_0(k) = \text{col}(\hat{x}_k, u_k)$.

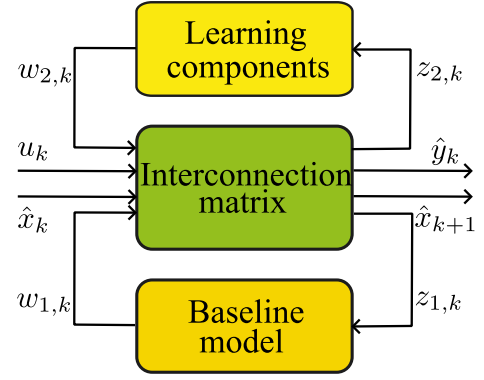


Fig. 1. LFR-based model augmentation of a baseline model.

can be readily represented, as well as a wide variety of structures beyond them. The following condition must be considered for these formulations.

Condition 1 (Well-posedness). For every state $\hat{x}_k$ and input u_k , model (4) admits a unique solution $z_k = [z_{1,k} \ z_{2,k}]$.

We enforce this condition by avoiding algebraic loops in the formulations of (4). This is equivalent to avoiding cycles in a graph as in the following theorem:

Theorem 1 (Acyclic Directed Graph Bang-Jensen & Gutin, 2008). The directed graph represented by the adjacency matrix S is acyclic if and only if it has a topological ordering.

The presence of a topological ordering can be computed in linear time (Bang-Jensen & Gutin, 2008), giving a computational methodology to check the well posedness condition for a given interconnection configuration.

3.3. Truncated loss function

In order to formulate the estimation of (3), the multiple-shooting-based objective function of SUBNET (Beintema, Schoukens et al., 2023) is adapted. This allows for the use of computationally efficient batch optimization methods popular in machine learning, while also increasing data efficiency (Beintema, Schoukens et al., 2023). The objective function consists of N subsections of length T of the available data D_N as shown in Fig. 2. This objective function is given as

$$V_{\text{trunc}}(\theta) = \frac{1}{2N(T+1)} \sum_{i=1}^N \sum_{\ell=0}^{T-1} \|\hat{y}_{k_i+\ell|k_i} - y_{k_i+\ell}\|_2^2 \quad (5a)$$

$$\begin{bmatrix} \hat{x}_{k_i+\ell+1|k_i} \\ \hat{y}_{k_i+\ell|k_i} \end{bmatrix} := \phi_\theta(\hat{x}_{k_i+\ell|k_i}, u_{k_i+\ell}) \quad (5b)$$

$$\hat{x}_{k|k} := \psi_\theta(y_{k-n_b}^{k-1}, u_{k-n_b}^{k-1}), \quad (5c)$$

where $\theta = [\theta_{\text{base}}^\top \ \theta_{\text{aug}}^\top \ \theta_{\text{encoder}}^\top]^\top$ is the joint parameter vector, $k+\ell|k$ indicates the state $\hat{x}_k$ or the output $\hat{y}_k$ at time $k+\ell$ simulated from the initial state $\hat{x}_{k|k}$ at time k ; and ϕ_θ is the LFR-based augmentation structure (4) which is constructed from ϕ_{base} , ϕ_{aug} and S . The subsections start at a randomly selected time $k_i \in \{n+1, \dots, N-T\}$. The initial state of these subsections is estimated by an encoder function ψ_θ from past input-output data, i.e., $\hat{x}_{k|k} = \psi_\theta(y_{k-n_b}^{k-1}, u_{k-n_b}^{k-1})$ where $u_{k-n_b}^{k-1} = [u_{k-n_b}^\top \ \dots \ u_{k-1}^\top]^\top$ for $\tau \geq 0$ and $y_{k-n_b}^{k-1}$ is defined similarly. This encoder is co-estimated with ϕ_θ .

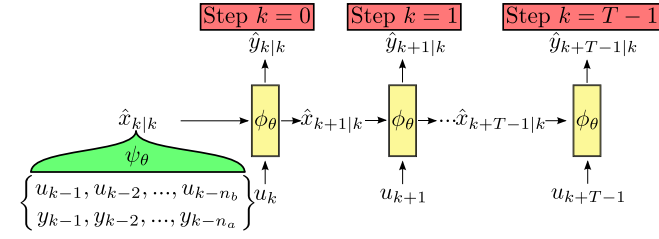


Fig. 2. SUBNET structure: the subspace encoder ψ_θ estimates the initial state at time index k based on past inputs and outputs, then the state is propagated through ϕ_θ multiple times until a simulation length T .

3.4. Regularized loss function

The LFR-based model (4) is overparameterized. The joint identification of (4) can therefore result in learning components canceling out part of the baseline model. As a result, the baseline model parameters θ_{base} can be significantly altered by the optimization process, leading to excessively large deviations from the initialized values θ_{base}^* . This can lead to a loss of interpretability of the baseline model. To address this issue, we adapt the regularization cost term from Bolderman (2024), which is given as

$$V_{\text{reg}}(\theta) = \left\| \Lambda (\theta_{\text{base}} - \theta_{\text{base}}^*) \right\|_2^2, \quad (6)$$

where Λ is a weighting matrix specifying the relative importance of each baseline model parameter. This matrix is computed as

$$\Lambda = \left(\frac{1}{\epsilon} V_{\text{MSE}}(\theta_{\text{base}}^*) \right)^{\frac{1}{2}} \text{diag}(\theta_{\text{base}}^*)^{-1}, \quad (7)$$

where V_{MSE} is the mean-square-error loss of the baseline model with parameters θ_{base}^* compared to the training data set, and ϵ is a tuning parameter specifying the permitted deviation from the initial parameter values. The final cost function with the regularization cost term becomes

$$V(\theta) = V_{\text{trunc}} + V_{\text{reg}}. \quad (8)$$

3.5. Baseline model normalization

For the estimation of ANNs, normalization of the input and output data to zero mean and to a standard deviation of 1 has shown to improve model estimation (Bishop, 1995). Therefore, the LFR-based model augmentation structure is normalized as in Beintema, Schoukens et al. (2023), while the learning components are initialized as in Schoukens and Tóth (2020). This means that $\hat{x}$, u , $\hat{y}$ in (4) are normalized. The to-be-augmented baseline model thus needs to have normalized input, state, and output as well. This is accomplished by the following transformation

$$\tilde{f}_{\text{base}} = T_x f_{\theta_{\text{base}}, \text{base}} (T_x^{-1} \tilde{x}, T_u^{-1} u) \quad (9a)$$

$$\tilde{h}_{\text{base}} = T_y h_{\theta_{\text{base}}, \text{base}} (T_x^{-1} \tilde{x}, T_u^{-1} u), \quad (9b)$$

where $T_u \in \mathbb{R}^{n_u}$ is a diagonal matrix of the inverse standard deviations σ_u^{-1} . The standard deviations σ_u are derived from D_N . T_y is similarly defined. For T_x , the standard deviation σ_x is determined in the baseline states of the model $\tilde{x}$. These states can be obtained by, e.g., simulation of the baseline model with nominal input and initial conditions.

3.6. Parameter estimation and properties

The encoder-based model augmentation method can readily be implemented with PyTorch. In particular, the use of the ADAM optimizer (Kingma, 2014) provides an efficient training method for the considered cost function and model structure. The original SUBNET

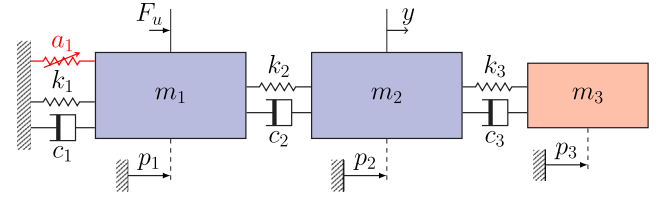


Fig. 3. 3-DOF MSD. The linear dynamics m_1 and m_2 are assumed known, while the dynamics of m_3 and the contribution of a_1 are aimed to be learnt from data.

Table 2

Physical parameters of the 3-DOF MSD.

Body	Mass m_i	Spring k_i	Damper c_i	Hardening a_i
1	0.5 kg	100 $\frac{\text{N}}{\text{m}}$	0.5 $\frac{\text{N s}}{\text{m}}$	100 $\frac{\text{N}}{\text{m}^3}$
2	0.4 kg	100 $\frac{\text{N}}{\text{m}}$	0.5 $\frac{\text{N s}}{\text{m}}$	–
3	0.1 kg	100 $\frac{\text{N}}{\text{m}}$	0.5 $\frac{\text{N s}}{\text{m}}$	–

method has been shown to provide consistency guarantees under the assumption of persistently exciting data and convergence of the optimization to the global minimum of the cost function (Beintema, Schoukens et al., 2023). Under the assumption that the system (1) is part of the model set spanned by the parameterization of (4), the consistency properties of the SUBNET method are inherited.

4. Mass spring damper simulation example

4.1. Mass-spring-damper system

As a first simulation example,² a 3 degrees-of-freedom (DOF) mass-spring-damper (MSD) is considered with a hardening spring nonlinearity, as shown in Fig. 3 with physical parameters as given in Table 2. The states associated with the system representation are the positions p_i and the accelerations $\dot{p}_i$ of the masses m_1 , m_2 , m_3 . The hardening spring is described as a cubic nonlinearity with parameter a_1 . The position p_2 is the measured output. MSD systems can be used to model a large variety of real world systems, e.g., structural (Lai et al., 2021), biomedical (Nikoooyan & Zadpoor, 2011) and mechatronic (Vaseur, 2017) systems. Thus, MSD systems are an interesting example to study.

We obtain the data for model estimation by applying 4th order Runge-Kutta (RK4) based numerical integration on the 3-DOF MSD system with $T_s = 0.02$ s. The system is simulated for a zero-order hold input signal F_u generated by a DT multisine u_k with 1666 frequency components in the range $[0, 25]$ Hz with a uniformly distributed phase in $[0, 2\pi)$. The sampled output measurements y_k are perturbed by additive noise e_k described by a DT white noise process with $e_k \sim \mathcal{N}(0, \sigma_e^2)$. Here, σ_e is chosen so that the signal-to-noise ratio (SNR) equals to 20, 30, or 60 dB. After removing the transient due to initial conditions, the generated sampled output y_k for the input u_k is collected in the data set D_N . Separate data sets are generated for estimation, validation, and testing with sizes $N_{\text{est}} = 2 \cdot 10^4$, $N_{\text{val}} = 10^4$, $N_{\text{test}} = 10^4$, respectively.

4.2. Baseline model

The baseline model is chosen to represent the linear 2-DOF MSD dynamics corresponding to the blue masses in Fig. 3, i.e., both the red mass m_3 and the nonlinear spring a_1 are left out. We consider two initializations for the baseline model parameters: the ideal values from Table 2, and approximate values from Table 3. The root mean squared simulation test error (RMSE) of the baseline model with the ideal parameters is shown in Table 4 and the approximate parameters in Table 5. Both models perform relatively poorly, despite (approximately) representing a large part of the system dynamics.

² <https://github.com/JanHHoekstra/Model-Augmentation-Public>.

Table 3

Approximate physical parameters for the 2-DOF MSD baseline model.

Body	Mass m_i	Spring k_i	Damper c_i
1	0.5 kg	95 $\frac{\text{N}}{\text{m}}$	0.45 $\frac{\text{N s}}{\text{m}}$
2	0.4 kg	95 $\frac{\text{N}}{\text{m}}$	0.45 $\frac{\text{N s}}{\text{m}}$

Table 4

RMS simulation test error of the baseline and estimated models for the 3-DOF MSD example with ideal parameter initialization for three SNR values.

Model	SNR 20 dB	SNR 30 dB	SNR 60 dB
Baseline	0.198	0.197	0.197
Static series	0.0162	0.00707	0.00255
Dynamic series	0.0153	0.00533	0.00172
Static parallel	0.0160	0.00673	0.00246
Linear dynamic parallel	0.0218	0.0161	0.0371
Dynamic parallel	0.0154	0.00554	0.00159
ANN-SS	0.0158	0.00572	0.00373

Table 5

RMS simulation test error of the estimated model augmentations for the 3-DOF MSD example with approximate parameter initialization for three SNR values.

Model	SNR 20 dB	SNR 30 dB	SNR 60 dB
Baseline	0.178	0.177	0.177
Static series	0.0163	0.00680	0.00257
Dynamic series	0.0152	0.00557	0.00106
Static parallel	0.0166	0.00661	0.00224
Linear dynamic parallel	0.0216	0.0160	0.0150
Dynamic parallel	0.0152	0.00577	0.00163

4.3. Estimated models

We estimate variations of the LFR-based augmentation structure³ corresponding to representations of the augmentation structures described in Table 1. For parallel augmentation structures, the learning components are chosen as feedforward neural networks, and for series augmentation structures, we choose residual networks (ResNet) (He et al., 2016). ResNets are chosen to capture the dominant linear behavior of the system in the series augmentation structure. This is not necessary for the parallel model, as the baseline model is additively augmented. For all learning components, the number of hidden layers and neurons are as in Table 6. The activation function is chosen as tanh for all models except the linear dynamic parallel model, which uses an identity activation function to represent a linear augmentation. For the dynamic augmentation, we add two additional states to the baseline model states for a total of 6 states. This is the minimum number of states required to model the 3-DOF MSD system. The baseline model, learning component, and encoder parameters are jointly estimated as described in Section 3. The hyperparameters for these estimations are shown in Table 6, with 16 nodes used for the encoder. The regularization tuning parameter ϵ in the joint identification cost function, was set at $\epsilon = 1$ for the ideal initialization and $\epsilon = 10^{12}$ for the approximate initialization. As a comparison to a black-box approach, an ANN-SS model parameterized by ResNets is estimated with the same estimation procedure and hyperparameters, but trained for 10 000 epochs.

The simulation RMSE on the test data for these estimated models with the ideal initialization are shown in Table 4 and for the approximate initialization in Table 5, both for three different levels of SNR. The dynamic augmentations are able to capture the model accurately, while the static augmentations result in slightly less accurate models. The linear dynamic augmentations result in significantly worse models.

³ The dynamic parallel model took 15 h to train on a 12th Gen Intel(R) Core(TM) i7-12700H. The other models took a similar duration of training. The authors note that the code was not optimized for runtime performance.

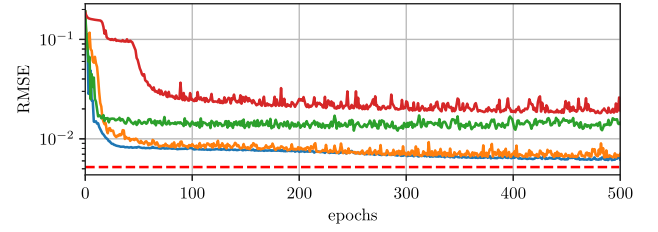


Fig. 4. Validation loss over first 500 training epochs for dynamic parallel (blue), static parallel (orange), linear dynamic parallel (green) and ANN-SS (red) models in the 3-DOF MSD example. The noise floor is shown with a dashed line.

Table 6

Hyperparameters for identifying the LFR-based augmentation and ANN-SS models for the 3-DOF MSD example.

Hidden layers	nodes	n_a n_y	n_a n_b	T	Epochs	Batch size
2	8	1	7	200	3000	2000

This indicates that the inclusion of additional states and nonlinear augmentations is beneficial for obtaining accurate augmentation models of the 3-DOF MSD system for the given baseline model.

ANN-SS models achieve RMSE scores similar to those of dynamic augmentations, as shown in Table 4. However, dynamic augmentations achieve this score in relatively few epochs, as shown in the validation loss curves in Fig. 4, while the ANN-SS models requires 10 000 epochs to reach an equivalent score. Table 4 also indicates that the identification algorithm performs well across multiple SNRs. However, for the 60 dB SNR, a model quality that is 100 times better than the 20 dB SNR case could be expected while only a factor 10 improvement is observed. This higher error is most likely caused by an incomplete convergence of the optimization as previous theoretical analysis of the SUBNET identification algorithm, on which the proposed method is built, has shown that the SUBNET estimator results in consistent estimates under mild conditions (Beintema et al., 2021).

Almost all estimated physical parameters remained close to the initialization values for both the ideal and approximate case. The value of c_1 moved to the ideal value for all approximate initialized models. Thus, the ideal initializations are not negated, but the approximate initializations are also not all corrected. This indicates that the learning components are learning part of the system dynamics that could be represented by the baseline model.

Further analysis of the dynamic parallel model is provided in Figs. 5 and 6. In Fig. 5, we show the simulation error on test data (blue) of the nonlinear dynamic parallel augmented model (orange). The figure shows consistent error performance across all test data. In Fig. 6, we show the comparison between the states $\hat{x}$ of the dynamic parallel model (blue) and the outputs of the learning components ϕ_{aug} (orange). Here, $\hat{x} = [x_1 \dots x_4]$ and $\bar{x} = [x_5 \ x_6]$. The output of the learning components is relatively small for $\bar{x}$, and $\bar{x}$ is modeled solely by the learning components. From this we can conclude that the learning components are augmenting the baseline model resulting in an accurate model and not replacing the baseline model with their own dynamics. Therefore, we have obtained an accurate model whose dominant behavior is given by an interpretable baseline model.

5. Bouc-Wen hysteresis simulation

We also consider the Bouc-Wen benchmark dataset (Schoukens & Noël, 2017). This dataset consists of sampled data of a 1-DOF MSD system with a nonlinear hysteretic spring. Two baseline models are considered starting from the linear 1-DOF MSD system ODE:

$$m\ddot{y}(t) + c\dot{y}(t) + ky(t) = u(t). \quad (10)$$

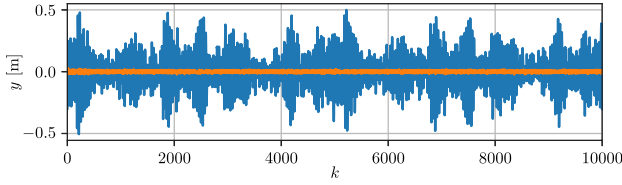


Fig. 5. Simulation error of the augmented model (orange) compared to the measured test data (blue) for the 3-DOF MSD example.

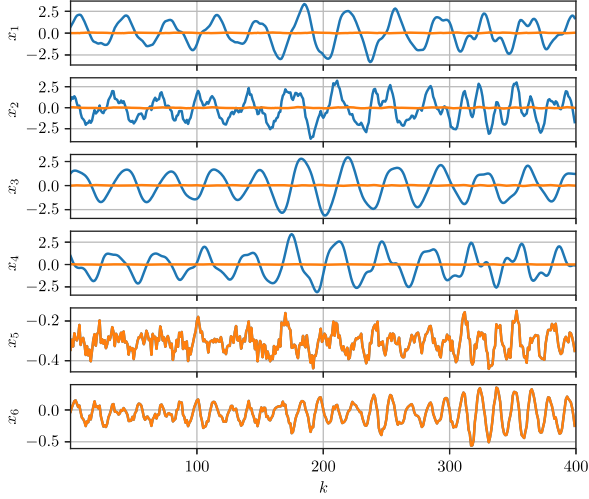


Fig. 6. Comparison of the baseline model states $\hat{x}$ (blue) and the outputs of the learning components ϕ_{aug} (orange) for a dynamic parallel augmentation structure in terms of the simulated model response on the test data. States $x_1 - x_4$ are the sum of the baseline model and the parallel augmentation contribution, while states $x_5 - x_6$ are the output of the dynamic augmentation only.

Firstly, a zero-order-hold discretized linear 1-DOF MSD system with its parameters set equal to the ideal values given in the benchmark (Schoukens & Noël, 2017) (see also Table 7), resulting in the following state-space equations:

$$x_{k+1} = \begin{bmatrix} 0.9560 & 0.0013 \\ -65.46 & 0.9494 \end{bmatrix} x_k + \begin{bmatrix} 4 \cdot 10^{-7} \\ 6.546 \cdot 10^{-2} \end{bmatrix} u_k, \quad (11)$$

$$y_k = \begin{bmatrix} 1 & 0 \end{bmatrix} x_k. \quad (12)$$

Secondly, a nonlinear FP-based model approximating the nonlinear dynamics of the Bouc-Wen system by an additional cubic spring term added to the discretized linear FP-based model is considered:

$$x_{k+1} = \begin{bmatrix} 0.9441 & 0.0013 \\ -82.32 & 0.8948 \end{bmatrix} x_k + \begin{bmatrix} 0 \\ 3.089 \cdot 10^5 \end{bmatrix} x_{1,k}^3 + \begin{bmatrix} 4 \cdot 10^{-7} \\ 6.374 \cdot 10^{-2} \end{bmatrix} u_k, \quad (13)$$

$$y_k = \begin{bmatrix} 1 & 0 \end{bmatrix} x_k, \quad (14)$$

where $x_{1,k}$ denotes the first state at time sample k . The coefficients (m , c , k and nonlinear spring constant k_3) of this nonlinear baseline model are estimated to obtain a best least-squares simulation performance on the training dataset. The resulting parameter values can be found in Table 7. Observe that the simulation error of the nonlinear baseline model is significantly lower than the linear one (see Table 9). This is due to a better estimate of the true damping in the system (the hysteretic term introduces extra damping that is not accounted for in the linear baseline model), and due to the addition of the cubic spring. However, the nonlinear baseline model still lacks the capability of capturing the full nonlinear hysteretic behavior of the system.

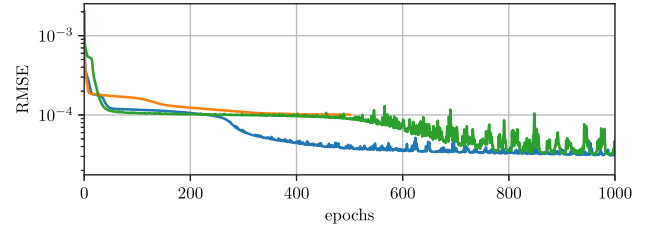


Fig. 7. Validation loss over first 1000 training epochs for dynamic parallel (blue), static parallel (orange) and ANN-SS (green) models.

Table 7

Parameter values of the Bouc-Wen baseline models.

	m	c	k	k_3
Linear	2	10	10^5	–
Nonlinear	2	77.42	129 160	$3.089 \cdot 10^5$

Table 8

Hyperparameters for identifying the LFR-based augmentation and ANN-SS models in the Bouc Wen example.

Hidden layers	Nodes	n_u	n_y	n_a	n_b	T	Epochs	Batch size
2	8	1	13	500	4000	2000		

Table 9

Root mean squared simulation error of the baseline and estimated models on the Bouc-Wen multisine test data.

Model	Linear FP	Nonlinear FP
Baseline	$170 \cdot 10^{-5}$	$19.5 \cdot 10^{-5}$
Static parallel	$9.76 \cdot 10^{-5}$	$9.48 \cdot 10^{-5}$
Dynamic parallel	$2.12 \cdot 10^{-5}$	$1.06 \cdot 10^{-5}$
ANN-SS	$2.79 \cdot 10^{-5}$	

Table 10

Root mean squared simulation error of the estimated models on the Bouc-Wen sinesweep test data.

Model	Linear FP	Nonlinear FP
Static parallel	$4.18 \cdot 10^{-5}$	$3.87 \cdot 10^{-5}$
Dynamic parallel	$1.21 \cdot 10^{-5}$	$1.03 \cdot 10^{-5}$
ANN-SS	$1.86 \cdot 10^{-5}$	

After discretization and transformation to a state-space form, we augment the baseline models with a static and dynamic parallel augmentation structure. The baseline model, learning component, and encoder parameters are jointly estimated as described in Section 3. The hyperparameters for these estimations are shown in Table 8, with 16 nodes used for the encoder. The regularization tuning parameter ϵ in the joint identification cost function, was set at $\epsilon = 1$. As a comparison to a black-box approach, an ANN-SS model parameterized by ResNets is estimated with the same estimation procedure and hyperparameters.

The RMSE scores of the baseline models as well as the augmentations are shown in Tables 9 and 10 for a multisine and sinesweep test respectively. The dynamic augmentation performs best in both cases, which is expected as the extra state is required to model the hysteretic dynamics. The state of the art performance on the sinesweep test also illustrates the good generalization performance on other input classes. In Fig. 7 we show the validation loss curves of the model augmentations and the ANN-SS model. Here we notice that the dynamic parallel augmentation converges fastest to a high-quality estimate.

In Fig. 8, we again show the comparison between the states $\hat{x}$ of the dynamic parallel model (blue) and the outputs of the learning components ϕ_{aug} (orange). The output of the learning components is still relatively small (but larger than the augmentation contributions for the 3-DOF MSD example) for $\hat{x}$, and $\bar{x}$ is modeled solely by the learning components.

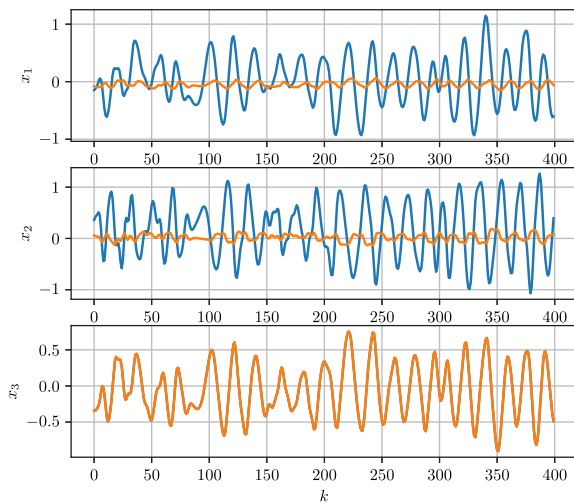


Fig. 8. Comparison of the baseline model states $\hat{x}$ (blue) and the outputs of the learning components ϕ_{aug} (orange) for a dynamic parallel augmentation structure in terms of the simulated model response on the test data using a linear baseline model. States x_1 - x_2 are the sum of the baseline model and the parallel augmentation, while state x_3 is the output of the dynamic augmentation only.

6. Conclusion

We have adapted an encoder-based estimation method for the augmentation of baseline models. For this, as an important contribution, we have introduced a flexible LFR-based model augmentation structure. We have shown how the proposed estimation method and LFR-based structure can be used for static and dynamic model augmentation in a 3-DOF MSD simulation study and on the Bouc-Wen benchmark dataset. The resulting models are accurate, estimated computationally efficiently, and retain the interpretability of the baseline model. We also observe that the baseline model parameters remain close to the initializations, preserving interpretability.

Fixed interconnections are considered in this work. Future research will investigate parameterized interconnections for use in automatic model structure selection.

CRedit authorship contribution statement

Jan H. Hoekstra: Methodology, Writing – review & editing, Data curation, Investigation, Validation, Conceptualization, Formal analysis, Software, Writing – original draft. **Chris Verhoek:** Conceptualization, Investigation, Methodology, Writing – review & editing. **Roland Tóth:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Resources, Supervision, Writing – review & editing. **Maarten Schoukens:** Investigation, Methodology, Project administration, Resources, Supervision, Validation, Writing – review & editing, Conceptualization, Data curation, Formal analysis, Funding acquisition.

Declaration of competing interest

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